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TOPICAL REVIEW

Electric Vehicles Under Low Temperatures: A Review on Battery Performance, Charging Needs, and Power Grid Impacts

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ABSTRACT Electric vehicles (EVs) are gaining mainstream adoption as more countries introduce net-zero carbon targets for the near future. Lithium-ion (Li-ion) batteries, the most commonly used energy storage technology in EVs, are temperature sensitive, and their performance degrades at low operating temperatures due to increased internal resistance. The existing literature on EV-power grid studies assumes that EVs are used under "perfect temperatures" (e.g. 21 Celsius) and temperature-related issues are ignored. In addition, most of the countries/regions with high EV penetration (e.g. Norway, Canada, northern parts of the US and China, etc.) experience harsh cold months, making it extremely critical to understand EV performance and consequently their impacts on the electrical power networks. In this paper, we present a systematic review of the literature that considers the combined investigation of Li-ion battery technology and power networks, focusing on their operation under suboptimal weather conditions. More specifically, we review: (i) the impact of low temperatures on the electrochemical performance of EV batteries in parking, charging and driving modes, (ii) the challenges experienced by EVs during charging and associated performance degradation, and (iii) the additional impacts of EV charging on the power networks. Our analysis shows that there are serious research gaps in literature and industry applications, which may hinder mass EV adoption and cause delays in charging station roll-out.

INDEX TERMS Electric vehicles, Li-ion battery, low temperatures, power grid impacts, power quality.

NOMENCLATURE

BMS	Battery Management System.
BTMS	Battery Thermal Management System.
CO_2	Carbon dioxide.
CC	Constant Current.
CV	Constant Voltage.
CENELEC	European Committee for Electrotechnical Standardization.
DSM	Demand Side Management.
DoD	Depth of Discharge.
EV	Electric Vehicle.
EVSE	Electrical Vehicle Supply Equipment.
EN	European Norm.
ESS	Energy Storage System.
FEC	Full Equivalent Cycles.
GHG	Greenhouse Gases.
HVAC	Heating, Ventilating and Air-Conditioning.
ICE	Internal Combustion Engine.
IEC	International Electrotechnical Commission.
Li-ion	Lithium-ion.
LV	Low Voltage.

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